

PYTHON PROJECT REPORT

AI PERSONAL ASSISTANT STIMULATION

A Command-Line Based Python Project

Project Title	AI Personal Assistant Stimulation
Programming Language	Python
Execution Environment	Command Line / Terminal
Project Type	Python Programming Project

Prepared as a Python course/project submission

1. Introduction

The AI Personal Assistant Stimulation is a Python-based command-line project that simulates basic personal-assistant functions. The program interacts with the user through text input and provides options for greeting, displaying the current time and date, telling a joke or poem, performing calculator operations, opening websites, accessing a weather search, calling a saved contact, viewing contacts, and exiting the assistant.

The project uses Python modules such as datetime and webbrowser together with variables, dictionaries, loops, conditional statements, operators, input/output, and string methods.

2. Objective

- To develop an interactive personal assistant simulation using Python.
- To provide several useful functions through a menu-driven interface.
- To practice input and output operations, conditions, loops, operators, and dictionaries.
- To use Python modules to obtain the current date/time and open web pages.
- To create a program that can be executed as a standalone Python file from the terminal.

3. Features

Feature	Implemented in the project
Greeting	Greets the user by name.
Time	Displays the current time using datetime.
Date	Displays the current date using datetime.
Joke	Displays a programmed computer joke.
Poem	Displays a short programmed poem.
Calculator	Accepts two numbers and an operator.
YouTube	Opens YouTube using webbrowser.
Google	Opens Google using webbrowser.
Weather	Opens a Google weather search using webbrowser.
Call a contact	Searches the saved contacts and opens a tel: link.
View contacts	Displays the saved contacts.
Exit	Displays a goodbye message and stops the loop.

4. Algorithm / Working

1. Start the program and import datetime and webbrowser.
2. Display the assistant title and ask the user for their name.
3. Create a contacts dictionary containing saved contact names and phone numbers.
4. Enter a while loop so that the assistant continues accepting commands.
5. Display the 12 available menu choices.
6. Read the user's choice.

7. Use if/elif conditions to execute the selected feature.
8. For the calculator, read two numbers and an operator and perform the selected operation.
9. For web functions, use webbrowser.open() to open the required website/search.
10. For calling, compare the entered contact name with the saved contacts using lower().
11. For viewing contacts, iterate through the dictionary using a for loop.
12. If the user selects 12, print the goodbye message and break the loop.
13. Stop.

5. Python Concepts Used

Concept	Use in this project
Variables	Store the user's name, menu choice, numbers, operator, date/time, and other values.
input() / print()	Used for interaction with the user and displaying results.
Dictionary	The contacts dictionary stores contact names and phone numbers.
while loop	Keeps the assistant menu running until Exit is selected.
for loop	Used to search contacts and display all saved contacts.
if / elif / else	Selects the operation corresponding to the user's menu choice.
Operators	Arithmetic operators are used by the calculator; comparison and logical conditions control decisions.
String methods	lower() is used for case-insensitive contact-name comparison.
datetime module	Used to obtain and format the current time and date.
webbrowser module	Used to open YouTube, Google, Weather search, and tel: links.

6. Important Code Sections — From the Submitted Project

The following excerpts are taken from the uploaded Jupyter HTML export of the project. They show the actual implementation rather than generic example code. The code is reproduced as implemented in the submitted project.

6.1 Imports and User Setup

```
import datetime
import webbrowser

print("=====")
print("          AI PERSONAL ASSISTANT          ")
print("=====")

name = input(" What is your name ?")
print("Hello", name + "!Iam your personal assistant.")
```

6.2 Saved Contacts and Main Menu Loop

```
contacts = {
    "Mom": "8072904701",
    "Dad": "9345260109",
    "Sister": "8925112755",
    "Brother": "7204371906",
    "Friend": "7373908686"
}

while True:
    print("\nWhat would you like to do ?")
    print("1. Greeting")
    print("2. Tell me the time")
```

```

print("3. Tell me the date")
print("4. Tell me a joke")
print("5. Tell me poem")
print("6. Open Calculator")
print("7. Open Youtube")
print("8. Open Google")
print("9. Open Weather")
print("10. Call a contact")
print("11. View contacts")
print("12. Exit")

choice = input("Enter your choice")

```

6.3 Time and Date Functions

```

elif choice == "2":
    current_time = datetime.datetime.now().strftime("%H:%M:%S ")
    print("Current Time :", current_time)

elif choice == "3":
    current_date = datetime.datetime.now().strftime("%d-%m-%y")
    print("Today's Date :", current_date)

```

6.4 Calculator Section

```

elif choice == "6":
    num1 = float(input("Enter first number:"))
    operator = input("Enter operator(+, -, *, /, %, **):")
    num2 = float(input("Enter second number:"))

    if operator == "+":
        print("Result =", num1 + num2)
    elif operator == "-":
        print("Result =", num1 - num2)
    elif operator == "*":
        print("Result =", num1 * num2)
    elif operator == "/":
        if num != 0:
            print("Result:", num1 % num2)
        else:
            print("Cannot divide by zero.")
    elif operator == "%":
        if num2 != 0:
            print("Result:", num1 % num2)
        else:
            print("Cannot divide by zero.")
    elif operator == "**":
        print("Result:", num1 ** num2)
    else:
        print("Invalid operator.")

```

6.5 Website Opening

```

elif choice == "7":
    print("Opening Youtube....")
    webbrowser.open("https://www.youtube.com")

elif choice == "8":
    print("Opening Google....")
    webbrowser.open("https://www.google.com")

elif choice == "9":
    print("Opening Weather....")
    webbrowser.open("https://www.google.com/search?q=weather")

```

6.6 Contact Search and Calling

```

elif choice == " 10 ":
    contact_name = input("Enter contact name :")
    found = False

    for contact in contacts:
        if contact.lower() == contact_name.lower():
            print("Calling", contact)
            print("Phone number:", contact[contact])
            webbrowser.open("tel:" + contacts[contact])
            found = True
            break

    if found == False:
        print("Contact not found.")

```

6.7 View Contacts and Exit

```
elif choice == "11":
    print("\nSaved Contacts:")
    for contact, number in contacts.items():
        print(contact, ":", number)

elif choice == "12":
    print("Goodbye", name + "!")
    break

else:
    print("Invalid choice. Please enter a number from 1 to 10.")
```

7. Sample Input and Output — Actual Project Output

The following output is taken from the uploaded project HTML export. It represents the actual outputs recorded from the project rather than invented sample results.

7.1 Program Start

```
=====
AI PERSONAL ASSISTANT
=====

Hello TANVE SRI!Iam your personal assistant.
```

7.2 Greeting

Hello TANVE SRI!How can i help you?

What would you like to do ?

1. Greeting
2. Tell me the time
3. Tell me the date
4. Tell me a joke
5. Tell me poem
6. Open Calculator
7. Open Youtube
8. Open Google
9. Open Weather
10. Call a contact
11. View contacts
12. Exit

7.3 Time

Current Time : 12:33:36

7.4 Date

Today's Date : 29-09-26

7.5 Joke

Why did the computer go the doctor ?
Because it had virus!

7.6 Poem

The darkest night will fade away ,
And lead you the brighter day.

7.7 Calculator

Result: 49787136.0

7.8 Website Features

Opening Youtube....
Opening Google....
Opening Weather....

7.9 View Contacts

Saved Contacts:
Mom : 8072904701
Dad : 9345260109
Sister : 8925112755

Brother : 7204371906
Friend : 7373908686

7.10 Exit

Goodbye TANVE SRI!

8. Execution and Project Structure

The project can be saved as a Python .py file and executed from a command prompt or terminal. The project uses the built-in datetime and webbrowser modules shown in the submitted code.

```
AI-Personal-Assistant/  
■  
■■■ ai_personal_assistant.py  
■■■ README.md  
■■■ report/  
    ■■■ AI_Personal_Assistant_Simulation_Project_Report.pdf
```

Run:
python ai_personal_assistant.py

9. Advantages and Limitations

Advantages: The project provides an interactive menu, combines several Python concepts in one application, uses standard Python modules, and can perform practical actions such as opening websites and handling saved contacts.

Limitations: The assistant is a programmed simulation. Its responses and available actions are determined by the conditions written in the program and are not equivalent to a full natural-language AI system.

10. Conclusion

The AI Personal Assistant Stimulation project demonstrates the practical use of Python programming concepts through an interactive assistant. The submitted implementation combines input/output, dictionaries, loops, conditional statements, operators, string methods, datetime, and webbrowser in a single application. The project provides a useful foundation for extending the assistant with additional commands and more advanced technologies in future work.

— End of Report —